Practical No: 6

Name: More Vaishnavi Dadaso

Roll No.: 269

Problem Statement: Design and implement the Heap Sort algorithm to efficiently sort an array of integers in ascending order. The implementation should be optimized for time and space complexity and should clearly demonstrate the working principles of heap data structures (min-heap or max-heap, as applicable)

Code:

```
# Heap Sort in Python


def heapify(arr, n, i):
  largest = i
  left = 2 * i + 1
  right = 2 * i + 2

  # Check left child
  if left < n and arr[left] > arr[largest]:
    largest = left

  # Check right child
  if right < n and arr[right] > arr[largest]:
    largest = right

  # If largest is not the root
  if largest != i:
    arr[i], arr[largest] = arr[largest], arr[i]

    # Recursively heapify the affected sub-tree
    heapify(arr, n, largest)
```

```
def heapSort(arr):
    n = len(arr)

    # Build Max Heap
    for i in range(n // 2 - 1, -1, -1):
        heapify(arr, n, i)

    # Heap Sort
    for i in range(n - 1, 0, -1):
        # Swap root (maximum) with last element
        arr[i], arr[0] = arr[0], arr[i]

        # Heapify the reduced heap
        heapify(arr, i, 0)

# ---------------- Main Program ---------------- #

arr = []

n = int(input("Enter Number of Elements: "))

for i in range(n):
    x = int(input("Enter Element: "))
    arr.append(x)

print("\nOriginal Array")
print(arr)

heapSort(arr)
```

```
print("\nSorted Array (Ascending)")
print(arr)
```

Output:

```
Enter Number of Elements: 8
Enter Element: 22
Enter Element: 78
Enter Element: 45
Enter Element: 34
Enter Element: 67
Enter Element: 60
Enter Element: 90
Enter Element: 65

Original Array
[22, 78, 45, 34, 67, 60, 90, 65]

Sorted Array (Ascending)
[22, 34, 45, 60, 65, 67, 78, 90]
```